

ronment. Solar/PV system design engineers deal with these issues on a day-to-day basis trying to make the output independent of environmental parameters.

Optimising power. A solar cell contains a host of power converters, viz, step-up, step-down, DC-DC and so on. Each converter has a specific function and they all need to be working optimally to ensure smooth running of the entire solar panel and, as such, the devices connected to it. Unless all the converters are designed in such a way that they all work and deliver performance on the expected lines, the design wouldn't be considered ideal for use.

Lowering production costs. The majority of the cost associated with solar cells is due to the land used for laying the panels. The trend is to increase the efficiency of these cells so as to get more energy from the same land.

Predicting manufacturing yields. Nowadays, with solar silicon wafers being sliced at thicknesses as small as 150 microns, wafer handling is becoming harder and harder as these dimensions make wafer and cell breakage much easier. The industry needs an in-line tool that rejects mechanically unstable silicon wafers before these are introduced into further cell processing and finds cracked cells before these are laminated into modules.

Solar/PV system design engineers work towards building a

system that is optimised for design performance and is reliable enough to achieve the desired target. They also aim to reduce the effects of varying parameters on the system performance while reducing the production costs, and hence a simulation-guided approach is the recommended way to go.

Saber for renewable energy systems

Saber allows you to operate in multiple domains together so as to best capture the essence of working conditions vis-a-vis solar/PV systems. It lets you enable full-system virtual prototyping, i.e., prototyping a product by utilising computer-aided design and computer-aided engineering software to validate a design before deciding to make a physical prototype.

With proper use of Saber, design engineers can solve the problem of active width optimisation by utilising the TCAD device design, physical parameters which contribute to interconnect resistances that can be extracted and a system-level model developed.

At cell level, Saber helps in maximising efficiency and optimising geometric and process parameters. At module level, it minimises the effect of interconnects on performance along with reducing the effect of cell variation on the performance of solar modules. At system level, the software maximises system performance by accounting for diurnal

solar inclination and tracking of the sun's path. System-level efficiency delivered to the grid, including the inverter system, is also maximised sufficiently.

To sum up

Today, system design requires reliability considerations to be taken into account early in the design process. In solar/PV design domain, this becomes even more important as the margin for error is quite thin in this field. Early adoption of a robust design philosophy through Saber ensures systematic implementation of design techniques that will lead to more reliable designs.

Saber provides a path from raw cell technology to an optimised power system (be it solar, hydel or wind) at a fraction of the traditional time and cost. It delivers the simulation capabilities required for virtual prototyping and validation of complex solar cell structures and power electronic systems. It ensures availability of tools required to simulate large signal behaviour of power converters, which are the backbone of a PV system.

By using the specifically designed Saber tool, optimum distributed power systems can be designed in a cost-effective way. Saber also helps in making the design robust, effective and efficient, thereby helping design engineers come up with a stronger product for the market. ●

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EFY Group's New Initiative

EB TIMES

Launched on Sept 11, 2012



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